Chapter 23

Three Dimensional Geometry

Only One Option Correct Type Questions

1. A line with positive direction cosines passes through the point $P(2, -1, 2)$ and makes equal angles with the coordinate axes. The line meets the plane $2x + y + z = 9$ at point Q. The length of the line segment PQ equals **[IIT-JEE-2009 (Paper-2)]**

   (A) 1 (B) $\sqrt{2}$

   (C) $\sqrt{3}$ (D) 2

2. If the distance of the point $P(1, -2, 1)$ from the plane $x + 2y - 2z = \alpha$, where $\alpha > 0$, is 5, then the foot of the perpendicular from P to the plane is **[IIT-JEE-2010 (Paper-2)]**

   (A) $\left(\frac{8}{3}, \frac{4}{3}, -\frac{7}{3}\right)$ (B) $\left(\frac{4}{3}, -\frac{4}{3}, \frac{1}{3}\right)$

   (C) $\left(\frac{1}{3}, \frac{2}{3}, \frac{10}{3}\right)$ (D) $\left(\frac{2}{3}, -\frac{1}{3}, \frac{5}{2}\right)$

3. The point P is the intersection of the straight line joining the points Q (2, 3, 5) and R (1, –1, 4) with the plane $5x - 4y - z = 1$. If S is the foot of the perpendicular drawn from the point T (2, 1, 4) to QR, then the length of the line segment PS is **[IIT-JEE-2012 (Paper-1)]**

   (A) $\frac{1}{\sqrt{2}}$ (B) $\sqrt{2}$

   (C) 2 (D) $2\sqrt{2}$

4. The equation of a plane passing through the line of intersection of the planes $x + 2y + 3z = 2$ and $x - y + z = 3$ and at a distance $\frac{2}{\sqrt{3}}$ from the point (3, 1, –1) is **[IIT-JEE-2012 (Paper-2)]**

   (A) $5x - 11y + z = 17$

   (B) $\sqrt{2}x + y = 3\sqrt{2} - 1$

   (C) $x + y + z = \sqrt{3}$

   (D) $x - 2\sqrt{y} = 1 - \sqrt{2}$

5. Perpendiculars are drawn from points on the line $\frac{x+2}{2} = \frac{y+1}{-1} = \frac{z}{3}$ to the plane $x + y + z = 3$. The feet of perpendiculars lie on the line **[JEE (Adv)-2013 (Paper-1)]**

   (A) $\frac{x}{5} = \frac{y-1}{8} = \frac{z-2}{-13}$ (B) $\frac{x}{2} = \frac{y-1}{3} = \frac{z-2}{-5}$

   (C) $\frac{x}{4} = \frac{y-1}{3} = \frac{z-2}{-7}$ (D) $\frac{x}{2} = \frac{y-1}{-7} = \frac{z-2}{5}$

6. Let P be the image of the point (3, 1, 7) with respect to the plane $x - y + z = 3$. Then the equation of the plane passing through P and containing the straight line $\frac{x}{1} = \frac{y}{2} = \frac{z}{1}$ is **[JEE (Adv)-2016 (Paper-2)]**

(A) $x + y - 3z = 0$
(B) $3x + z = 0$
(C) $x - 4y + 7z = 0$
(D) $2x - y = 0$

7. The equation of the plane passing through the point (1, 1, 1) and perpendicular to the planes $2x + y - 2z = 5$ and $3x - 6y - 2z = 7$, is **[JEE (Adv)-2017 (Paper-2)]**

(A) $-14x + 2y + 15z = 3$
(B) $14x - 2y + 15z = 27$
(C) $14x + 2y - 15z = 1$
(D) $14x + 2y + 15z = 31$

One or More Option(s) Correct Type Questions

8. If the straight lines $\frac{x-1}{2} = \frac{y+1}{k} = \frac{z}{2}$ and $\frac{x+1}{5} = \frac{y+1}{2} = \frac{z}{k}$ are coplanar, then the plane(s) containing these two lines is (are) **[IIT-JEE-2012 (Paper-2)]**

(A) $y + 2z = -1$
(B) $y + z = -1$
(C) $y - z = -1$
(D) $y - 2z = -1$

9. A line ℓ passing through the origin is perpendicular to the lines

$l_1 : (3+t)\hat{i} + (-1+2t)\hat{j} + (4+2t)\hat{k}, \ -\infty < t < \infty$

$l_2 : (3+2s)\hat{i} + (3+2s)\hat{j} + (2+s)\hat{k}, \ -\infty < s < \infty$

Then, the coordinate(s) of the point(s) on ℓ_2 at a distance of $\sqrt{17}$ from the point of intersection of ℓ and ℓ_1 is (are) **[JEE (Adv)-2013 (Paper-1)]**

(A) $\left(\frac{7}{3}, \frac{7}{3}, \frac{5}{3}\right)$
(B) (–1, –1, 0)
(C) (1, 1, 1)
(D) $\left(\frac{7}{9}, \frac{7}{9}, \frac{8}{9}\right)$

10. Two lines $L_1 : x = 5, \frac{y}{3-\alpha} = \frac{z}{-2}$ and $L_2 : x = \alpha, \frac{y}{-1} = \frac{z}{2-\alpha}$ are coplanar. Then α can take value(s) **[JEE (Adv)-2013 (Paper-2)]**

(A) 1
(B) 2
(C) 3
(D) 4

11. From a point $P(\lambda, \lambda, \lambda)$, perpendiculars PQ and PR are drawn respectively on the lines $y = x, z = 1$ and $y = -x, z = -1$. If P is such that $\angle QPR$ is a right angle, then the possible value(s) of λ is(are) **[JEE (Adv)-2014 (Paper-1)]**

(A) $\sqrt{2}$
(B) 1
(C) –1
(D) $-\sqrt{2}$

12. In $\mathbb{R}^3$, consider the planes $P_1 : y = 0$ and $P_2 : x + z = 1$. Let P_3 be a plane, different from P_1 and P_2, which passes through the intersection of P_1 and P_2. If the distance of the point (0, 1, 0) from P_3 is 1 and the distance of a point (α, β, γ) from P_3 is 2, then which of the following relations is (are) true? **[JEE (Adv)-2015 (Paper-1)]**

(A) $2\alpha + \beta + 2\gamma + 2 = 0$
(B) $2\alpha - \beta + 2\gamma + 4 = 0$
(C) $2\alpha + \beta - 2\gamma - 10 = 0$
(D) $2\alpha - \beta + 2\gamma - 8 = 0$

13. In $\mathbb{R}^3$, let L be a straight line passing through the origin. Suppose that all the points on L are at a constant distance from the two planes $P_1 : x + 2y - z + 1 = 0$ and $P_2 : 2x - y + z - 1 = 0$. Let M be the locus of the feet of the perpendiculars drawn from the points on L to the plane P_1. Which of the following points lie(s) on M? **[JEE (Adv)-2015 (Paper-1)]**

(A) $\left(0, -\frac{5}{6}, -\frac{2}{3}\right)$ (B) $\left(-\frac{1}{6}, -\frac{1}{3}, \frac{1}{6}\right)$

(C) $\left(-\frac{5}{6}, 0, \frac{1}{6}\right)$ (D) $\left(-\frac{1}{3}, 0, \frac{2}{3}\right)$

14. Consider a pyramid $OPQRS$ located in the first octant ($x \geq 0$, $y \geq 0$, $z \geq 0$) with O as origin, and OP and OR along the x-axis and the y-axis, respectively. The base $OPQR$ of the pyramid is a square with $OP = 3$. The point S is directly above the mid-point T of diagonal OQ such that $TS = 3$. Then **[JEE (Adv)-2016 (Paper-1)]**

(A) The acute angle between OQ and OS is $\frac{\pi}{3}$

(B) The equation of the plane containing the triangle OQS is $x - y = 0$

(C) The length of the perpendicular from P to the plane containing the triangle OQS is $\frac{3}{\sqrt{2}}$

(D) The perpendicular distance from O to the straight line containing RS is $\sqrt{\frac{15}{2}}$

15. Let $P_1 : 2x + y - z = 3$ and $P_2 : x + 2y + z = 2$ be two planes. Then, which of the following statements(s) is (are) TRUE? **[JEE (Adv)-2018 (Paper-1)]**

(A) The line of intersection of P_1 and P_2 has direction ratios 1, 2, –1

(B) The line $\frac{3x-4}{9} = \frac{1-3y}{9} = \frac{z}{3}$ is perpendicular to the line of intersection of P_1 and P_2

(C) The acute angle between P_1 and P_2 is 60°

(D) If P_3 is the plane passing through the point (4, 2, –2) and perpendicular to the line of intersection of P_1 and P_2, then the distance of the point (2, 1, 1) from the plane P_3 is $\frac{2}{\sqrt{3}}$

16. Let L_1 and L_2 be the following straight lines. $L_1 : \frac{x-1}{1} = \frac{y}{-1} = \frac{z-1}{3}$ and $L_2 : \frac{x-1}{-3} - \frac{y}{-1} = \frac{z-1}{1}$. Suppose the straight line $L : \frac{x-\alpha}{l} = \frac{y-1}{m} = \frac{z-\gamma}{-2}$ lies in the plane containing L_1 and L_2, and passes through the point of intersection of L_1 and L_2. If the line L bisects the acute angle between the lines L_1 and L_2, then which of the following statements is/are TRUE? **[JEE (Adv)-2020 (Paper-1)]**

(A) $\alpha - \gamma = 3$ (B) $l + m = 2$

(C) $\alpha - \gamma = 1$ (D) $l + m = 0$

17. Let $\alpha, \beta, \gamma, \delta$ be real numbers such that $\alpha^2 + \beta^2 + \gamma^2 \neq 0$ and $\alpha + \gamma = 1$. Suppose the point (3, 2, –1) is the mirror image of the point (1, 0, –1) with respect to the plane $\alpha x + \beta y + \gamma z = \delta$. Then which of the following statements is/are TRUE? **[JEE (Adv)-2020 (Paper-2)]**

(A) $\alpha + \beta = 2$ (B) $\delta - \gamma = 3$

(C) $\delta + \beta = 4$ (D) $\alpha + \beta + \gamma = \delta$

18. Let P_1 and P_2 be two planes given by **[JEE (Adv)-2022 (Paper-1)]**

P_1: $10x + 15y + 12z – 60 = 0$,

P_2: $–2x + 5y + 4z – 20 = 0$.

Which of the following straight lines can be an edge of some tetrahedron whose two faces lie on P_1 and P_2?

(A) $\frac{x-1}{0}=\frac{y-1}{0}=\frac{z-1}{5}$ (B) $\frac{x-6}{-5}=\frac{y}{2}=\frac{z}{3}$

(C) $\frac{x}{-2}=\frac{y-4}{5}=\frac{z}{4}$ (D) $\frac{x}{1}=\frac{y-4}{-2}=\frac{z}{3}$

19. Let S be the reflection of a point Q with respect to the plane given by **[JEE (Adv)-2022 (Paper-1)]**

$\vec{r}=-(t+p)\hat{i}+t\hat{j}+(1+p)\hat{k}$

where t, p are real parameters and $\hat{i}, \hat{j}, \hat{k}$ are the unit vectors along the three positive coordinate axes. If the position vectors of Q and S are $10\hat{i}+15\hat{j}+20\hat{k}$ and $\alpha\hat{i}+\beta\hat{j}+\gamma\hat{k}$ respectively, then which of the following is/are TRUE?

(A) $3(\alpha + \beta) = –101$ (B) $3(\beta + \gamma) = –71$

(C) $3(\gamma + \alpha) = –86$ (D) $3(\alpha + \beta + \gamma) = –121$

Linked Comprehension Type Questions

Paragraph for Q.Nos. 20 to 21

Consider the lines :

$L_1: \frac{x+1}{3}=\frac{y+2}{1}=\frac{z+1}{2}$

$L_2: \frac{x-2}{1}=\frac{y+2}{2}=\frac{z-3}{3}$ **[IIT-JEE-2008 (Paper-2)]**

Choose the correct answer :

20. The unit vector perpendicular to both L_1 and L_2 is

(A) $\frac{-\hat{i}+7\hat{j}+7\hat{k}}{\sqrt{99}}$ (B) $\frac{-\hat{i}-7\hat{j}+5\hat{k}}{5\sqrt{3}}$

(C) $\frac{-\hat{i}+7\hat{j}+5\hat{k}}{5\sqrt{3}}$ (D) $\frac{7\hat{i}-7\hat{j}-\hat{k}}{\sqrt{99}}$

21. The shortest distance between L_1 and L_2 is

(A) 0 (B) $\frac{17}{\sqrt{3}}$

(C) $\frac{41}{5\sqrt{3}}$ (D) $\frac{17}{5\sqrt{3}}$

22. The distance of the point (1, 1, 1) from the plane passing through the point (–1, –2, –1) and whose normal is perpendicular to both the lines L_1 and L_2 is

(A) $\frac{2}{\sqrt{75}}$ (B) $\frac{7}{\sqrt{75}}$

(C) $\frac{13}{\sqrt{75}}$ (D) $\frac{23}{\sqrt{75}}$

Matrix-Match / Matrix Based Type Questions

23. Consider the following linear equations

$ax + by + cz = 0$

$bx + cy + az = 0$

$cx + ay + bz = 0$

Match the conditions/expression in **Column-I** with statements in **Column-II** and indicate your answer by darkening the appropriate bubbles in the 4 × 4 matrix given in the ORS. **[IIT-JEE-2007 (Paper-1)]**

Column-I	Column-II
(A) $a + b + c \neq 0$ and $a^2 + b^2 + c^2 = ab + bc + ca$	(p) the equations represent planes meeting only at a single point
(B) $a + b + c = 0$ and $a^2 + b^2 + c^2 \neq ab + bc + ca$	(q) the equations represent the line $x = y = z$
(C) $a + b + c \neq 0$ and $a^2 + b^2 + c^2 \neq ab + bc + ca$	(r) the equations represent identical planes
(D) $a + b + c = 0$ and $a^2 + b^2 + c^2 = ab + bc + ca$	(s) the equations represent the whole of the three dimensional space

24. Consider the lines $L_1 : \frac{x-1}{2} = \frac{y}{-1} = \frac{z+3}{1}$, $L_2 : \frac{x-4}{1} = \frac{y+3}{1} = \frac{z+3}{2}$ and the planes $P_1 : 7x + y + 2z = 3$, $P_2 = 3x + 5y - 6z = 4$. Let $ax + by + cz = d$ be the equation of the plane passing through the point of intersection of lines L_1 and L_2, and perpendicular to planes P_1 and P_2.

Match List-I with List-II and select the correct answer using the code given below the lists:

[JEE (Adv)-2013 (Paper-2)]

List-I	List-II
P. a =	1. 13
Q. b =	2. –3
R. c =	3. 1
S. d =	4. –2

Codes :

	P	Q	R	S
(A)	3	2	4	1
(B)	1	3	4	2
(C)	3	2	1	4
(D)	2	4	1	3

Integer / Numerical Value Type Questions

25. If the distance between the plane Ax – 2y + z = d and the plane containing the lines $\frac{x-1}{2}=\frac{y-2}{3}=\frac{z-3}{4}$ and $\frac{x-2}{3}=\frac{y-3}{4}=\frac{z-4}{5}$ is $\sqrt{6}$, then $|d|$ is **[IIT-JEE-2010 (Paper-1)]**

26. Let P be a point in the first octant, whose image Q in the plane $x + y = 3$ (that is, the line segment PQ is perpendicular to the plane $x + y = 3$ and the mid-point of PQ lies in the plane $x + y = 3$) lies on the z-axis. Let the distance of P from the x-axis be 5. If R is the image of P in the xy-plane, then the length of PR is _____. **[JEE (Adv)-2018 (Paper-2)]**

Question Stem for Question No. 27

Question Stem

Let α, β and γ be real numbers such that the system of linear equations

$x + 2y + 3z = \alpha$

$4x + 5y + 6z = \beta$

$7x + 8y + 9z = \gamma - 1$

is consistent. Let $|M|$ represent the determinant of the matrix

$$M = \begin{bmatrix} \alpha & 2 & \gamma \\ \beta & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$$

Let P be the plane containing all those (α, β, γ) for which the above system of linear equations is consistent, and D be the **square** of the distance of the point (0, 1, 0) from the plane P.

27. The value of D is _____. **[JEE (Adv)-2021 (Paper-1)]**

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